



Fig. 7: Construction of polymer thick-film flex circuit

flexible circuits, but there is significant variation between the different types in terms of their construction. Following is a review of the most common types of flexible circuit constructions. Flexible circuits have a rich history and are extremely diverse in their nature. This diversity opens them to use in a wide range of applications, with new applications being developed on a regular basis.

It is hard to predict where the technology will go next. However, roll-to-roll processing is likely to play an important part. Flexible electronics opens the door to foldaway smartphone displays, solar cells on a roll of plastic and advanced medical devices—if we can figure out how to make them.

Single-sided flex circuits

A single-layer flex circuit is the most basic and consists of a flexible polyimide film laminated to a thin sheet of copper. The copper layer is then chemically etched to produce a circuit pattern specific to your design requirements. A polyimide cover layer is then added for insulation and environmental protection of the circuit.

A single-sided flex is used for:

1. Dynamic flexing applications
2. Unusual folding and forming applications
3. Installation/service applications/repair
4. When there are limitations on space/thickness

Some important features of single-sided flex are:

1. Very thin construction, under 0.1mm-0.2mm (0.004-0.008 inch)
2. One conductor layer
3. Reverse-bared or back-bared pads
4. Supported and unsupported finger areas

Back-bared flex circuits

Double access flex, also known as back-bared flex, are flexible circuits having a single conductor layer but they allow access to selected features of the conductor pattern from both sides. While this type of circuit has certain benefits, the specialised processing requirements for accessing the features limit its use.

A dual-access flex circuit is simply a single-sided flex circuit that is manufactured such that its conductive material can be accessed from both sides of the circuit. Flexible Circuit is a front runner in this technology, which utilises specialised lasers and processing to skive open the polyimide layer of the base material to allow dual access to the single copper layer. While this type of circuit has certain benefits, the specialised processing requirements for accessing the features limit its use.

These flex circuits find application in such fields as:

1. Medical
2. Environmental
3. Test and Measurement
4. Automotive and other general electronic devices

Sculptured flex circuits

Sculptured flex circuits are a novel subset of normal flexible circuit structures. Their manufacturing process involves a special flex circuit multi-step etching method to produce a flexible circuit having finished copper conductors whose thickness differs at various places along their length. The conductors are generally thin in flexible areas and thick at interconnection points.

Sculptured circuit technology enables the thickness of the copper conductors to vary at any point on the circuit. In addition, selective ap-

plication of the supporting dielectric enables integral exposed fingers to be produced. All holes are etched, providing the flexibility to create precisely-located apertures of any shape or size at any position on the circuit.

Sculptured circuits are in use across a wide spectrum of applications from motor sports to missiles. Sculptured circuits, when combined with flexible or rigid circuits, provide an extremely cost-effective solution to complex interconnect problems by reducing cost, simplifying assembly and increasing reliability.

Sculptured flex circuits offer many benefits, including robust contact areas, nil or low tooling cost, reliability and cost-effectiveness. Their applications include power circuits and custom-built circuits.

Double-sided flex circuits

Double-sided flex circuits are flex circuits having two conductor layers. These can be fabricated with or without plated-through holes, though the plated-through-hole variation is much more common. When constructed without plated-through holes, the connection features are accessed from one side only, and the circuit is called Type V (5), according to military specifications. It is not a common practice but it is an option.

Because of the plated-through hole, terminations for electronic components are provided for on both sides of the circuit, thus allowing components to be placed on either side. Depending on design requirements, double-sided flex circuits can be fabricated with protective cover layers on one, both or neither side of the completed circuit. Generally, these are produced with the protective layer on both sides.

One major advantage of this type of substrate is that, it allows crossover connections to be made very easily. Many single-sided circuits are built on a double-sided substrate, just because they have one or two crossover connections. An example of this use is the circuit connecting a mouse pad to the motherboard of a laptop. All connec-